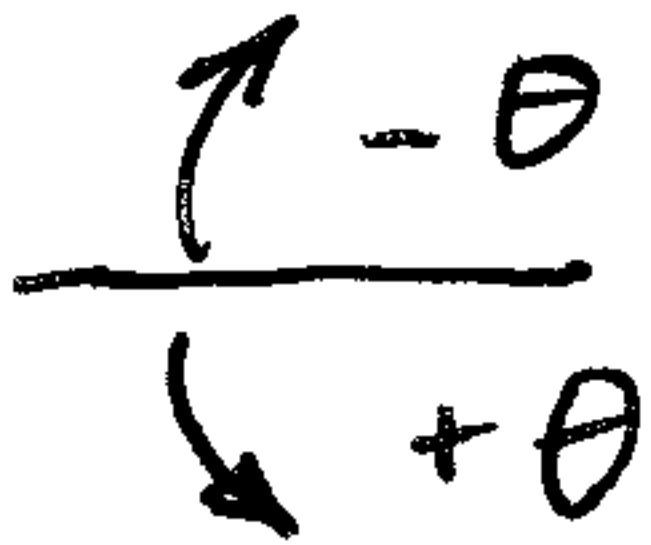
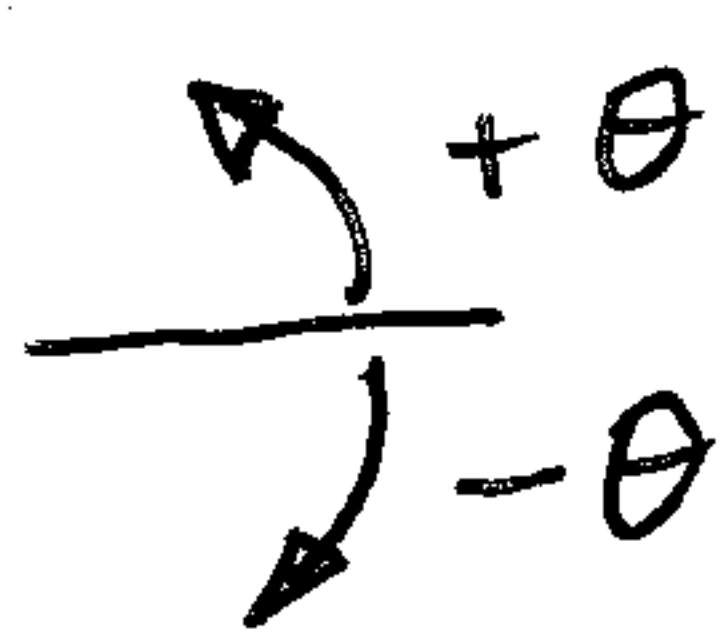


The θ_p Equation always gives the angle
From the I_x orientation To the nearest
 principal I (I_{MAX} or I_{MIN}). It also gives
 the correct sense or direction:



$$CW = -\theta$$

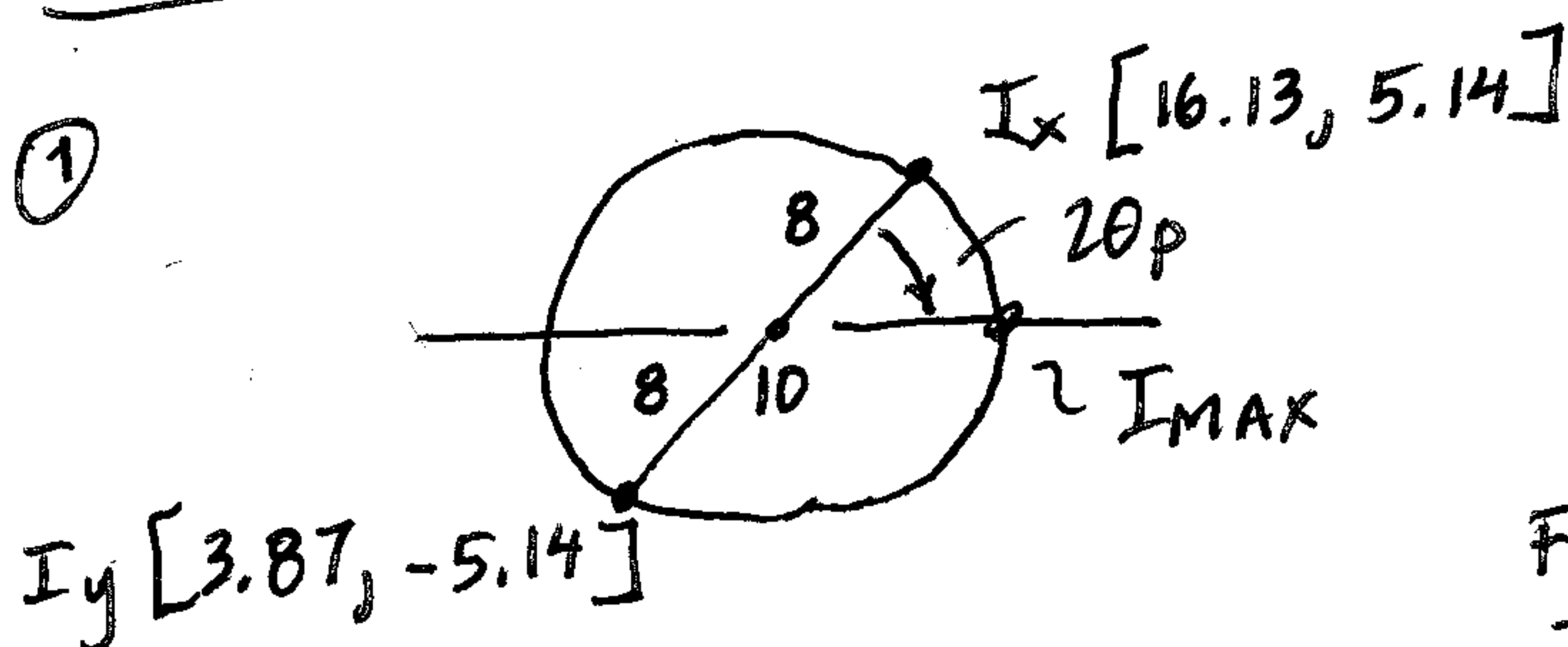
$$CCW = +\theta$$

θ_{p1} is angle
 from x
 to I_{MAX}
 θ_{p2} is angle
 to I_{MIN}

The θ_p equation:

$$\theta_p = \frac{1}{2} \tan^{-1} \frac{-I_{xy}}{\left(\frac{I_x - I_y}{2}\right)}$$

Four cases (made up) [$I_{AVG} = 10$, $R = 8$]

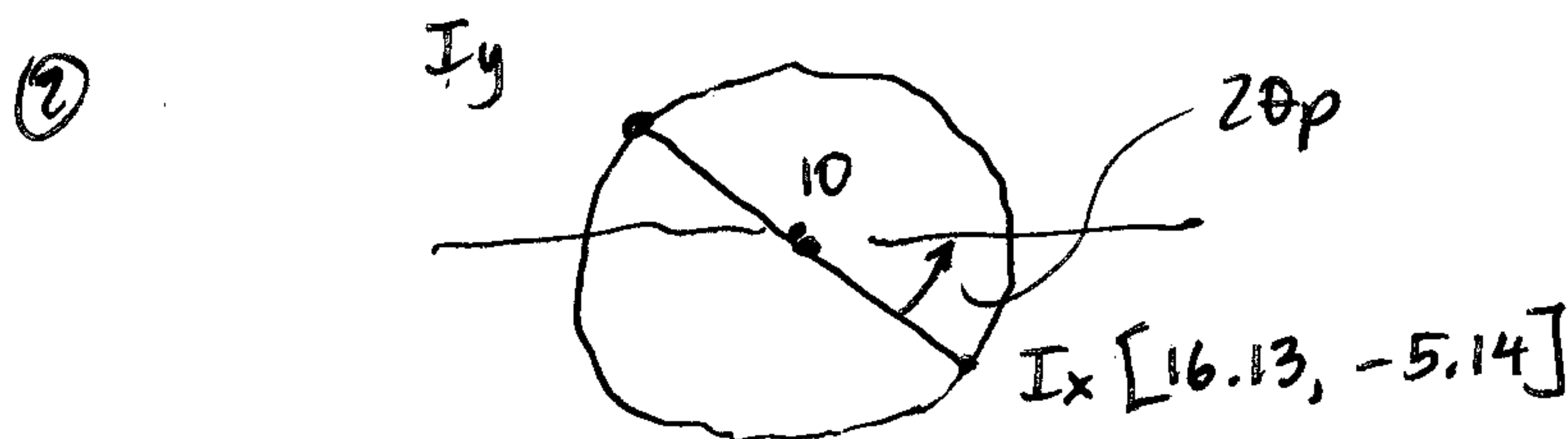


Calculate:

$$2\theta_p \text{ (on MC)} = -40^\circ$$

$$\theta_p \text{ (on Area)} = -20^\circ$$

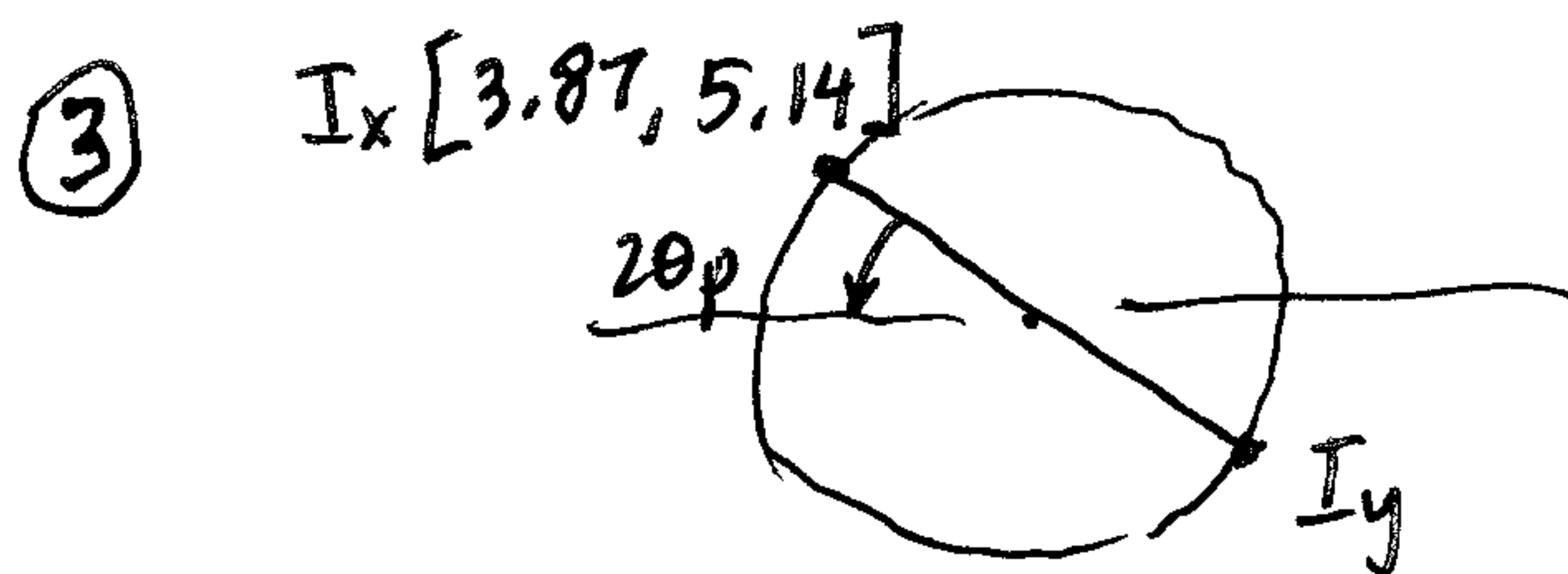
From I_x to I_{MAX} (This is θ_{p1})



$$2\theta_p = +40^\circ$$

$$\theta_p = +20^\circ$$

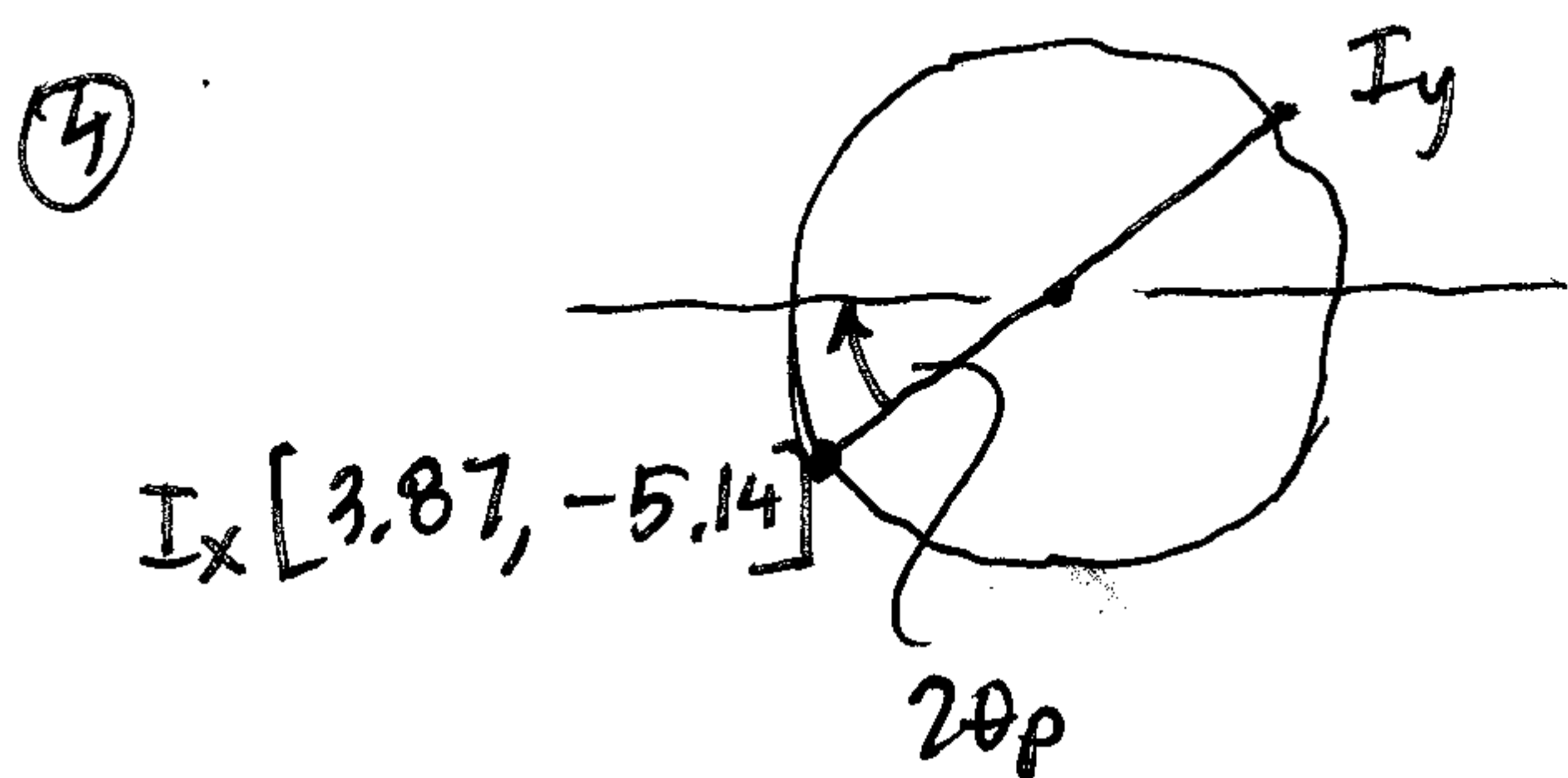
To I_{MAX} . (θ_{p1})



$$2\theta_p = +40^\circ$$

$$\theta_p = +20^\circ$$

To I_{MIN} . (θ_{p2})



$$2\theta_p = -40^\circ$$

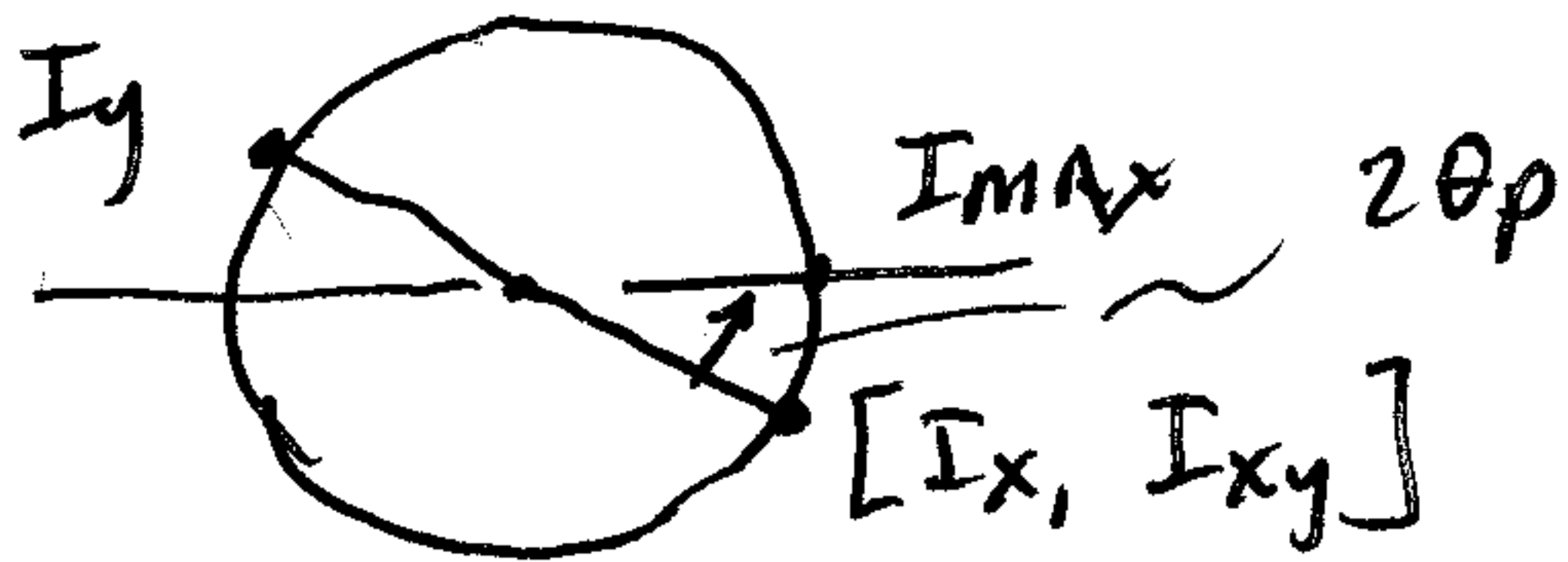
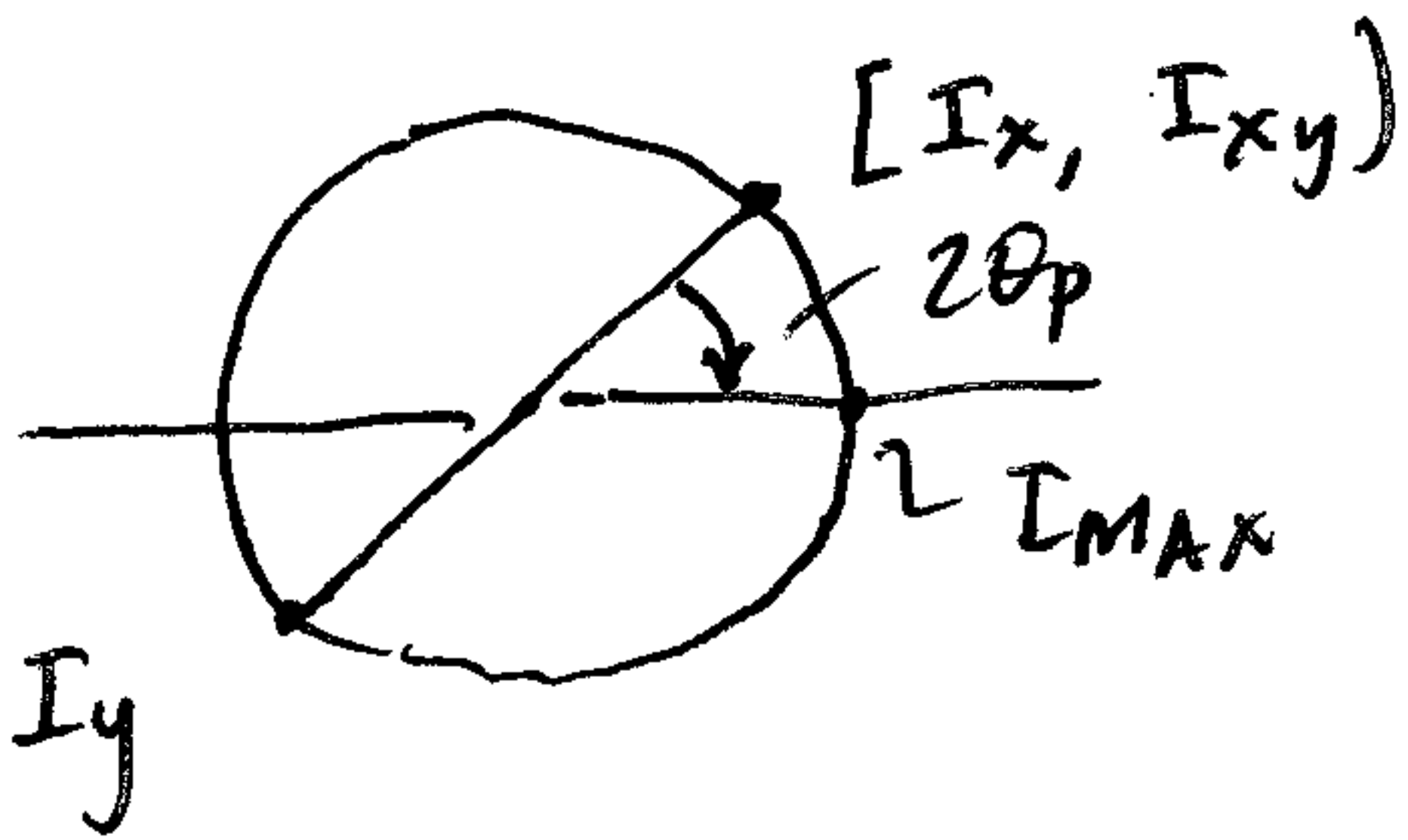
$$\theta_p = -20^\circ$$

To I_{MIN} . (θ_{p2})

Is θ_p to I_{MAX} or to I_{MIN} ?

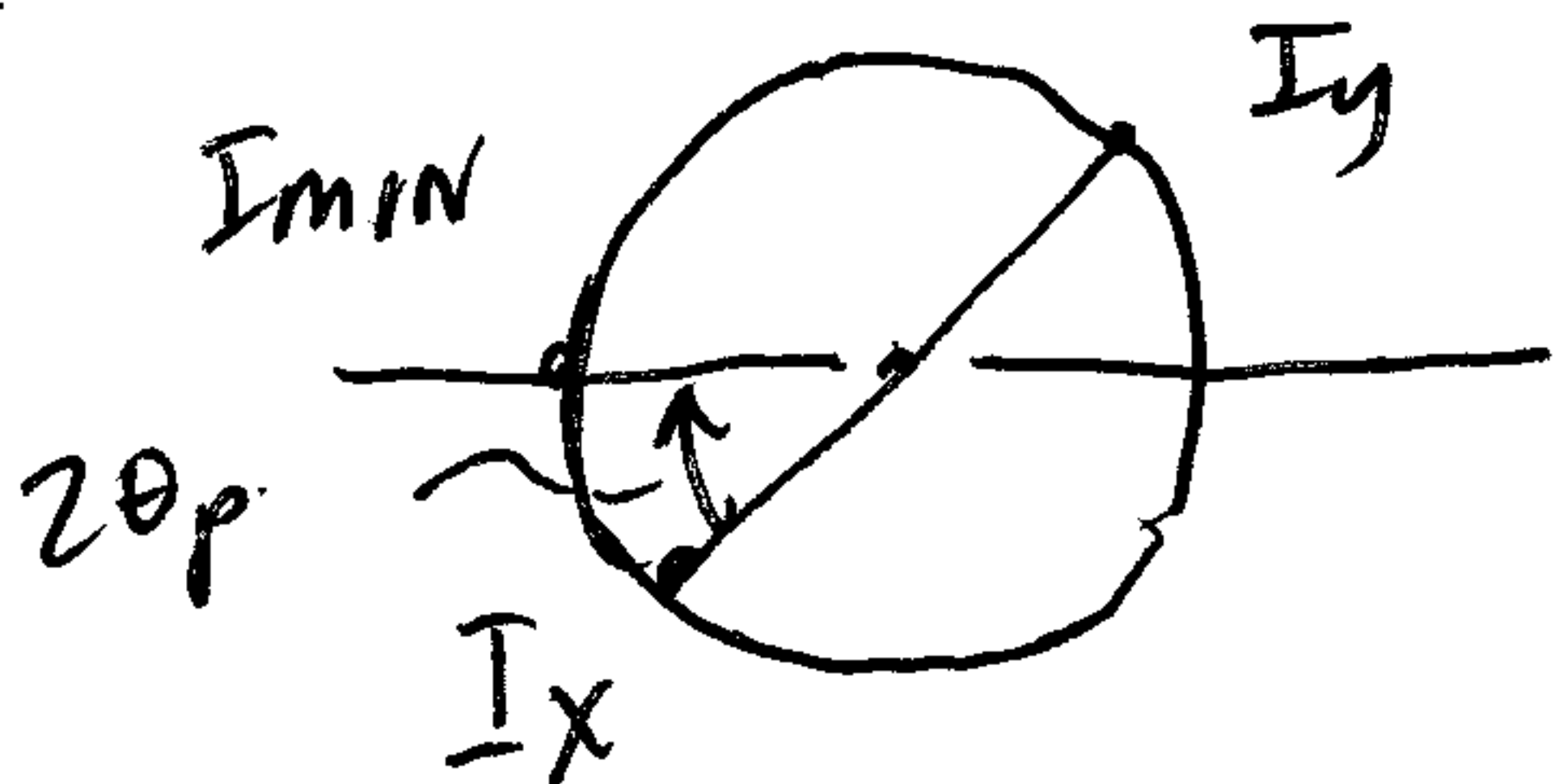
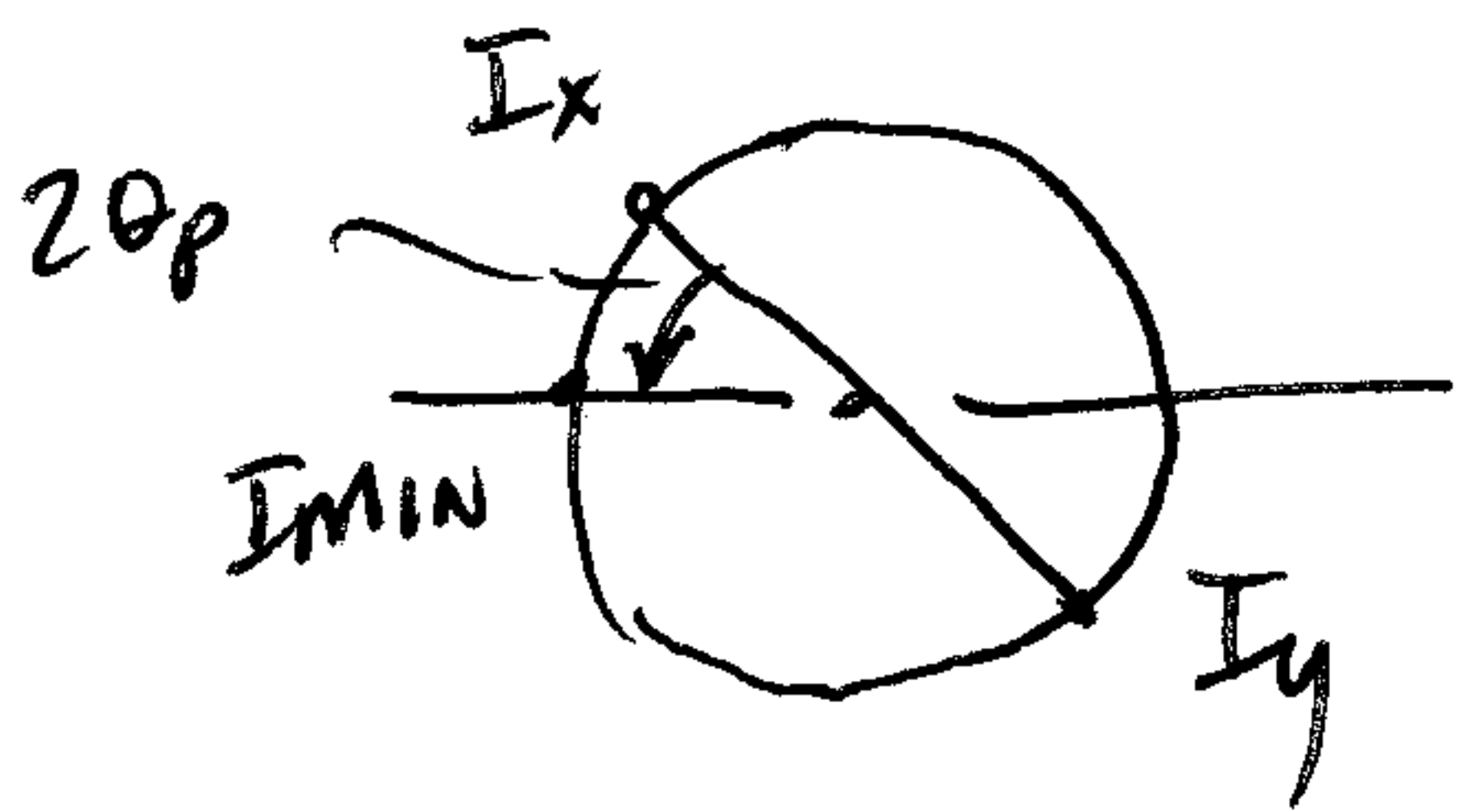
In other words, is it θ_{p1} (to I_{MAX})?
or θ_{p2} (to I_{MIN})?

If $I_x > I_y$, then $\theta_p = \theta_{p1}$ to I_{MAX} :



Since θ_p is
always measured
from I_x to
the nearest
principal MOI.

If $I_x < I_y$, then $\theta_p = \theta_{p2}$ to I_{MIN} .



For all of these, use
same equation:

$$\tan 2\theta_p = \frac{-I_{xy}}{\left(\frac{I_x - I_y}{2}\right)}$$